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20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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Abstract

Table of Contents

Abstract I
Introduction 1
Theory/Literature Review 2
Hypotheses 2
    Hypothesis 1 2
    Hypothesis 2 3
Analysis 3
    Data Cleaning 4
    Results 4
        Smaller-party representation across elections 4
        Relative smaller-party representation across elections 5
        Interpretation of the Results 6
Discussion 6
Conclusion 6
Bibliography 6
    Textsources 6
    Datasources 7
Appendix 10
    Appendix 1: Explanation Data Cleaning Process 10
    Appendix 2: Code for Data Cleaning 10
    Declaration of AI-Assistance 10

Introduction

On the 18th of December 2002 the Swiss Federal Supreme Court (FSC) came to a conclusion, which disrupted the Swiss Electoral System. In BGE 129 | 185 the FSC criticized that in the elections for the Council of the City of Zurich, the thresholds for parties to win a seat, vary heavily from district to district. Through that not every voter has the same influence and therefore is the Principle of Equal Treatment (Gleichhandlungsgebot) not given anymore. Based on this criticism the court decided that the City of Zurich needs to implement corrective measures, it let the City decide which these would be. (BGE 129 I 185, E. 7.3, S. 199–200) The Government of the Canton of Zurich decided to keep the consituency structure (Regierungsrat des Kantons Zürich, 2014, pp. 3–4). Instead the canton of Zurich comissioned the german mathematician Friedrich Pukelsheim to develop a comprehensive and at the same time minimally invasive solution for the problem. Eventually Pukelsheim came up with a solution named "biproportional divisor method with standard rounding" which is today often simply called "double Pukelsheim" or "Biproportional apportionment". (Lüscher, 2024, pp. 9-10)

Since the implementation through the Canton of Zurich in 2007, nine more cantons followed: Schaffhausen (2008), Aargau (2009), Nidwalden (2014), Zug (2014), Schwyz (2016) , Valais (2017) , Uri (2020), Graubünden (2022) in 2025 also the voters from Basel Landschaft decided to implement the Biproportional apportionment from 2027 on (Lüscher , 2024, p.17 ; Kanton Basel-Landschaft, 2025a, p.7 ; Kanton Basel-Landschaft, 2025b)

Switzerland's cantonal electoral reforms of the last two decades have brought renewed attention to a central question of representative democracy: how parliamentary seats can be distributed in a way that reflects voter preferences as fairly as possible while still preserving territorial representation. In the Swiss case, this debate became especially prominent because traditional proportional systems in small constituencies often generated high implicit thresholds. These electoral systems therefore dis-advantaged smaller parties which led to raising concerns about electoral equality (Bundeskanzlei, 2013, pp. 8–10). The double-proportional allocation method, commonly known as the double Pukelsheim, emerged in this context as a reform intended to reduce such distortions and to strength-en the equal weight of votes across the canton (Lüscher, 2023, p. 571).

At the core of this debate lies the contrast between two different logics of seat allocation. The traditional Hagenbach-Bischoff method belongs to the family of divisor methods with downward round-ing and therefore tends to benefit larger parties, especially in smaller electoral districts. (Bun-deskanzlei, 2013, pp. 6–7). By contrast, the double Pukelsheim combines a canton-wide upper alloca-tion with a subsequent redistribution of seats to districts, thereby improving proportionality at the cantonal level and reducing the number of votes that remain without effective influence on seat allo-cation (Pukelsheim & Schuhmacher, 2011, pp. 1581–1582, 1591). In this sense, the reform is closely linked to the principle of Erfolgswertgleichheit, meaning that votes should contribute as equally as possible to the final electoral outcome (Bundeskanzlei, 2013, pp. 8–9).

This reform is especially relevant for the representation of smaller parties. Because the double Pukelsheim aggregates party strength across the entire canton before allocating seats to districts, votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level (Lüscher, 2025, pp. 1–2). Sandro Lüscher illustrates this mechanical effect with the case of Schwyz in 2016: under the old Hagenbach-

Bischoff system, the Green Liberal Party would not have won a single seat, whereas under the new method they obtained three seats (Lüscher, 2025, p. 2-3). More generally, recent work suggests that the shift away from Hagenbach-Bischoff re-duces the structural advantage previously enjoyed by larger parties and improves the parliamentary chances of smaller competitors (Bundeskanzlei, 2025, pp. 22–25). At the same time, the reform does not eliminate all tensions. While the double Pukelsheim improves party proportionality and vote equality, it may also produce new frictions at the district level. Schol-ars have pointed out that the system can weaken the direct correspondence between a district's seat share and its actual influence on the canton-wide outcome, thereby creating a tension between parti-san proportionality and territorial representation (Lüscher, 2023, pp. 577–580).

Against this background, this thesis examines the mechanical effect of electoral reform on the parlia-mentary representation of smaller parties. More specifically, it asks and shows statistcally how many seats smaller parties gain through the transition from the Hagenbach-Bischoff system to the double Pukelsheim in Swiss cantonal parliamentary elections.

The central research gap of this thesis lies in the fact that the existing literature mainly discusses the Doppelproporz in legal, normative, or system-level terms: it emphasizes improved proportionality, reduced weightless votes, and better opportunities for smaller parties, while also pointing to trade-offs such as territorial distortions and counterintuitive seat allocations. However, there is still little systematic evidence on the mechanical seat effect of the reform itself: namely, how many seats small-er parties actually gained or would have lost when comparing the observed Pukelsheim allocation with a counterfactual Hagenbach-Bischoff allocation across elections and cantons. Recent work shows that smaller parties were generally better represented under the new system and that research has increasingly focused on broader institutional effects or territorial representation, but not on a stand-ardized empirical measurement of seat gains for smaller parties across cases. The thesis fills this gap by translating the reform's abstract proportionality advantage into a concrete, comparable outcome measure: the net seat gain or loss of smaller parties under the new versus the old allocation rule.

Theory/Literature Review

Hypotheses

To be able to answer the research question: 20 Years of Double Pukelsheim in Switzerland: What Is the Mechanical Effect of the Transition from Hagenbach-Bischoff on the Seat Allocation of Smaller Parties and Party-System Fragmentation?

We need to formulate hypotheses. The hypotheses are based on the theoretical considerations and the literature review. The first hypothesis focuses on the mechanical effect of the electoral reform on smaller parties, while the second hypothesis addresses the potential impact of the reform on party-system fragmentation.

Hypothesis 1

For each election e, smaller parties are those with less than 10% of the canton-wide vote. Let S_e^{DP} and S_e^{HB} be their combined seats under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation, respectively, and let M_e be the total number of seats in that election. Their gain in seat share, shown for each election in Figure 2, is

$$\Delta_e = 100\frac{S_e^{DP} - S_e^{HB}}{M_e} \quad \text{percentage points .}$$

The main hypothesis concerns the unweighted mean of these election-level gains, $\bar{\Delta} = E^{-1}\sum_{e=1}^{E}\Delta_e$, across the E analyzed elections. Thus, "more than 5% more seats" means a mean gain greater than **five percentage points of all available seats**, rather than a 5% increase relative to the Hagenbach-Bischoff seat total.

H_0 (null hypothesis): The mean gain in smaller parties' seat share under Double Pukelsheim is at most five percentage points: $\bar{\Delta} \leq 5$.

H_1 (main hypothesis): The mean gain in smaller parties' seat share under Double Pukelsheim exceeds five percentage points: $\bar{\Delta} > 5$.

The 10% threshold is based on that the Swiss Federal Supreme Court has argued that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed 10% (ADD SOURCE HERE)

Hypothesis 2

H_02 (null hypothesis): The transition to the double Pukelsheim system did not lead to a systematic increase in party-system fragmentation.

$$H_{0,2}: \quad \mathrm{ENP}_{\mathrm{post}} - \mathrm{ENP}_{\mathrm{pre}} \leq 0$$

H_12: The transition to the double Pukelsheim system led to a systematic increase in party-system fragmentation.

$$H_{1,2}: \quad \mathrm{ENP}_{\mathrm{post}} - \mathrm{ENP}_{\mathrm{pre}} > 0$$

$$\mathrm{ENP}_v = \frac{1}{\sum_{i=1}^{n} v_i^2} \qquad \mathrm{ENP}_s = \frac{1}{\sum_{i=1}^{n} s_i^2}$$

v = votes shares, s = seat shares

Analysis

H1: The analysis takes the actual seat allocation under the Pukelsheimer Proporz as its starting point. Based on the available election data at the municipal or district level, it then reconstructs how many seats each party would have obtained under the former Hagenbach-Bischoff system. The comparison between the actual seat distribution under the new system and the hypothetical distribution under the old system makes it possible to identify how many seats each party gained or lost as a result of the reform. In this way, the study assesses the mechanical effect of the electoral-system change on party representation, with a particular focus on smaller parties. For the purposes of this thesis, a smaller party is defined as a party with less than 10% of the canton-wide vote share in the relevant election. Youth lists and other special-interest lists, such as wom-en's lists or pensioners' lists, are not treated as independent parties. Instead, their votes and seats are attributed to the respective mother party, which reflects common electoral practice in Switzerland. (ADD SOURCE HERE) The choice of 10% as the threshold is grounded in both legal and normative considerations. In its de-cision BGE 131 I 74, the Swiss Federal Supreme Court argued that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed

10%. The 2013 report of the Federal Chancellery refers to the same benchmark and further notes that the Venice Commission considers thresholds in the range of 10–15% to be the upper acceptable limit . Using vote share rather than absolute party size is particularly appropriate because a party may be electorally small in one region while being comparatively strong in another. Low thresholds are closely linked to the principle of Erfolgswertgleichheit, or equality of vote success value, which requires that votes should be translated into seats as equally and effectively as possible. Investigating the threshold is especially important, since none of the reform cantons had a <10% threshold in all districts before the implantation of the new electoral reform (Lüscher, 2024, p. 115).

Data Cleaning

To be able to analyze the data from the different elections and compare the outcomes. each dataset has to be cleaned. The data For some cantons e.g. Valais and Schwyz the data is very scattered, therefore for there multiple datasets per election have to be cleaned and merged.

The procedure for the data cleaning was the following: For each canton in alphabetically order, for each election the data was cleaned.

To ensure the compatibility of the different datasets, the column names were standardized. The column names were renamed for each dataset to the following: “total_votes”, “svp_votes”, “sp_votes”, “cvp_votes”, “fdp_votes”, “glp_votes”, “bdp_votes”, “green_votes”, “other_votes”. It might be that some of the datasets do not contain all of these columns, since the party landscapes varies geographically and over time.

For the counterfactual, the district-level party votes and district seat magnitudes from the pre-Pukelsheim era are retained. The Hagenbach-Bischoff allocation follows the relevant cantonal law of the time (e.g., Sections 12-13 of the Aargau Grossratswahlgesetz for the 2005 election). The main counterfactual treats every party independently, ignoring list alliances, as they were abolished with the transition to Double Pukelsheim. Vote shares use each canton’s *Wählerzahl* definition: district-level party votes are divided by the district seat magnitude and rounded to the nearest integer before cantonal aggregation.

The official Double Pukelsheim results (`seats_dp`) are sourced from official cantonal authorities and are documented in the Bibliography. A single lookup table consolidates all official results for easy merging with the Hagenbach-Bischoff counterfactuals. All data processing is implemented in canton-specific R scripts (R/canton_*.R) which apply shared functions for algorithm execution and validation.

Results

Smaller-party representation across elections

Figure 1 compares the observed Double Pukelsheim allocation with the counterfactual Hagenbach-Bischoff allocation for every cleaned election. For each election, the figure aggregates the seats and canton-wide vote shares of all parties below the 10% threshold defined above. The grey line connects the two allocations derived from the same election.

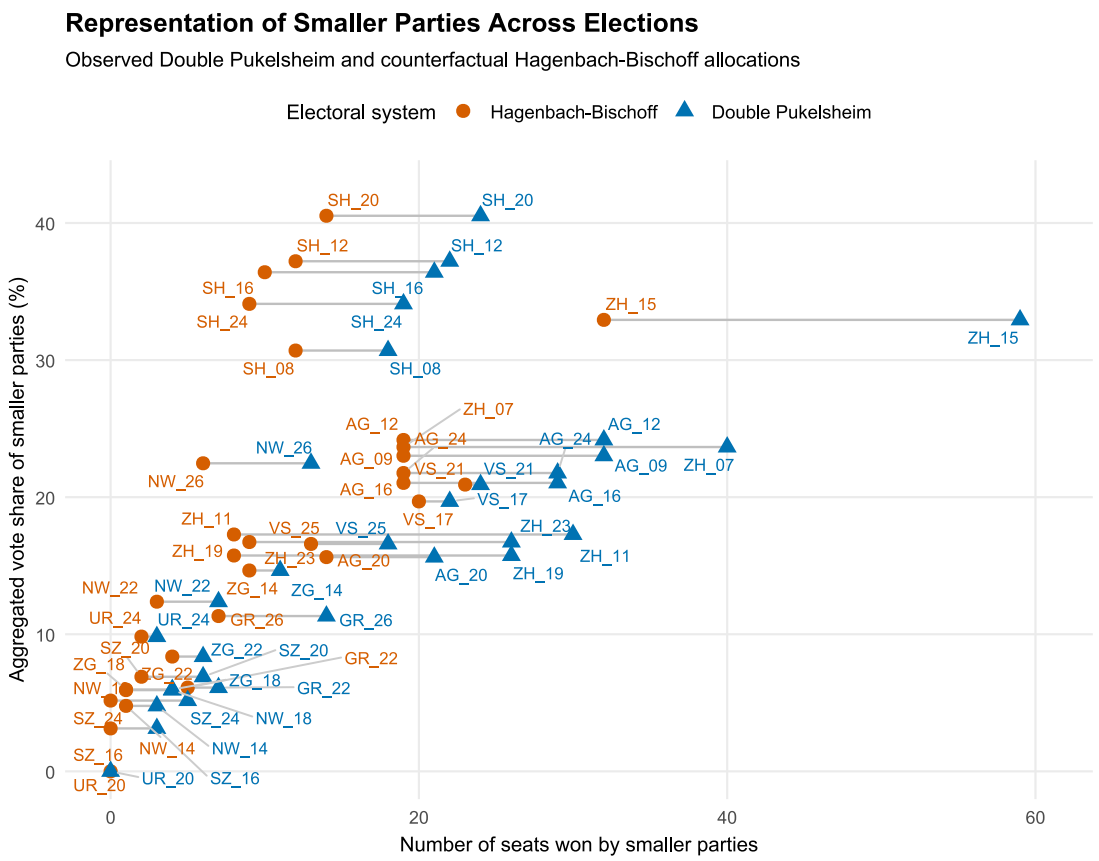


Figure 1: Seats won by smaller parties under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation. Smaller parties are those with less than 10% of the canton-wide vote share.

Relative smaller-party representation across elections

Figure 2 shows the same elections using the percentage of all parliamentary seats won by smaller parties on the x-axis. Each label reports the gain under Double Pukelsheim as a percentage-point share of all available seats.

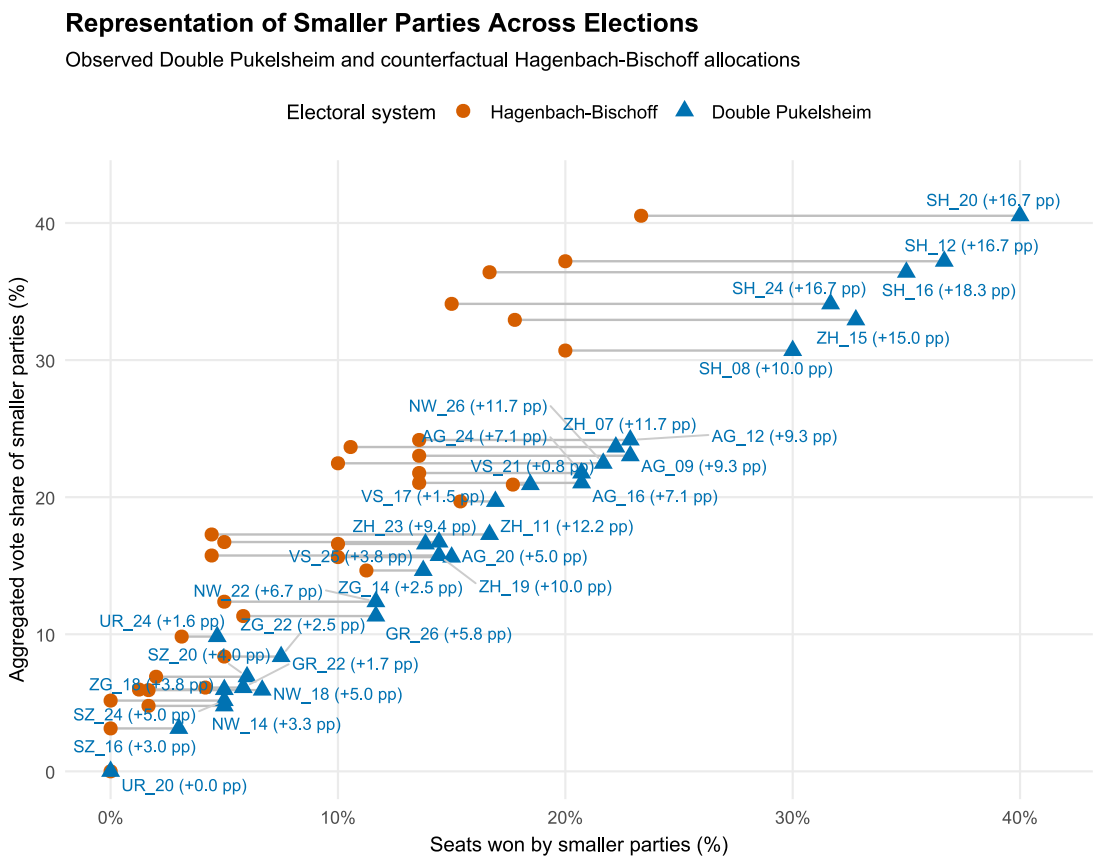


Figure 2: Percentage of seats won by smaller parties under Double Pukelsheim and counterfactual Hagenbach-Bischoff.

> **i** Hypothesis 1 summary
>
> Across all 32 elections, the unweighted mean gain in smaller parties' share of all seats was 7.4 percentage points (Q1: 3.2; median: 6.2; Q3: 10.4). The mean weighted by each election's parliament size was 7.5 percentage points. 17 individual elections exceeded five percentage points.

Interpretation of the Results

Discussion

Conclusion

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Appendix

Appendix 1: Explanation Data Cleaning Process

Appendix 2: Code for Data Cleaning

Declaration of AI-Assistance